

STM32CubeIDE can be used to create applications for STM32 devices using STM32Cube libraries (both HAL and Low Layer versions)

Prerequisites:

- B-U585I-IOT02A STM board
- USB cable

To start developing for the B-U585I-IOT02A board you will need to **create a new account with ST and then to download** the following software from STM:

- STM32CubeMX <https://www.st.com/en/development-tools/stm32cubemx.html>
- STM32CubeIDE <https://www.st.com/en/development-tools/stm32cubeide.html>
- STM32CubeProgrammer <https://www.st.com/en/development-tools/stm32cubeprog.html>

Click “Get Latest” for your platform of choice and “Accept” to accept the agreement terms. In order to download the necessary software, you will need to create an account with STM so please do it first.

Extract the zip files and begin the installation of each software in turn (first STM32CubeMX, then STM32CubeIDE and finally STM32CubeProgrammer). If an error occurs before the installation starts, check the permissions of you Antivirus (i.e. in Trellix disable Threat prevention->Access protection) and of Windows. You can choose to install the software only for your user to avoid requiring administrator access.

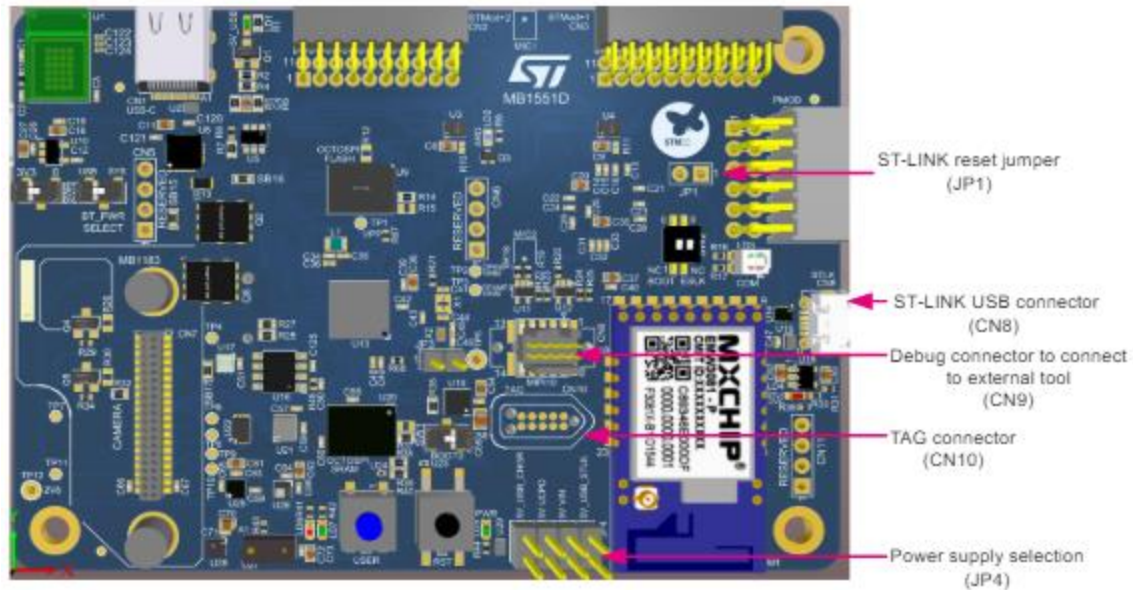
Essential documents:

- STM32U585AI datasheet <https://www.st.com/resource/en/datasheet/stm32u585ai.pdf>
- PM0264 - STM32 Cortex-M33 MCUs programming manual
https://www.st.com/resource/en/programming_manual/pm0264-stm32-cortexm33-mcus-programming-manual-stmicroelectronics.pdf
- Arm Cortex-M33 Processor Technical Reference Manual
<https://developer.arm.com/documentation/100230/latest/>
- UM2839 Discovery kit for IoT node with STM32U5 series
https://www.st.com/resource/en/user_manual/um2839-discovery-kit-for-iot-node-with-stm32u5-series-stmicroelectronics.pdf

To program and debug the onboard STM32 MCU:

We will use the STLINK-V3E programming and debugging tool integrated into the B-U585I-IOT02A Discovery kit. Check that the SW5 switch is in the OFF position and SW4 in the ON position (on the back of the board) to make sure that the ST-LINK is connected to the target MCU. This should already be the case, so unless you modified your board, you are already good to go! You can also consult Figure 27 and Figure 28 in the UM2839 Discovery kit for IoT node with STM32U5 series document (listed above) *if something is not working with your board*, to make sure that all the jumpers/switches are in the correct position.

- 1) **Before connecting the Discovery kit for the IoT node board to a Windows PC via USB**, the user must install a driver for the STLINK-V3E. It is available on <https://www.st.com/en/development-tools/stsw-link009.html> In case the B-U585I-IOT02A Discovery kit board is connected to the PC before the driver is installed, some B-U585I-IOT02A Discovery kit interfaces may be declared as “Unknown” in the PC device manager. In this case, you must install the dedicated driver files, and update the driver of the connected device from the device manager. Download the drivers, unzip them and execute dpinst_amd64.exe
- 2) Connect a USB cable to the ST-LINK USB connector (CN8) and from there to your PC:



- 3) Open STM32CubeMX. Click Help -> Manage embedded software packages -> scroll to STM32U5 and select the latest version -> Install
- 4) From STM32CubeMX, select File -> New Project -> Board Selector -> B-U585I-IOT02A -> Start Project. Answer **Yes** to the Initialize all peripherals with their default mode? Popup and **without TrustZone activated** in the second popup.
- 5) In the pinout view, try to locate PH6 and PH7 (hint: you can use the search function on the bottom right of the screen. What are they associated with?

- 8) In the Code Generator tab, ensure that the following options are checked:
- In STM32Cube Firmware Library Package section: Copy only the necessary library files
 - In Generated files section: Keep user code when regenerating the C code option, which only applies to the user sections within the STM32CubeMX generated files.

Project	STM32Cube MCU packages and embedded software packs ☐ Copy all used libraries into the project folder ☒ Copy only the necessary library files ☐ Add necessary library files as reference in the toolchain project configuration file
Code Generator	Generated files ☐ Generate peripheral initialization as a pair of '.c/.h' files per peripheral ☐ Backup previously generated files when re-generating ☒ Keep User Code when re-generating ☒ Delete previously generated files when not re-generated
Advanced Settings	HAL Settings ☐ Set all free pins as analog (to optimize the power consumption) ☐ Enable Full Assert
	User Actions Before Code Generation Browse After Code Generation Browse
	Template Settings Select a template to generate customized code Settings...

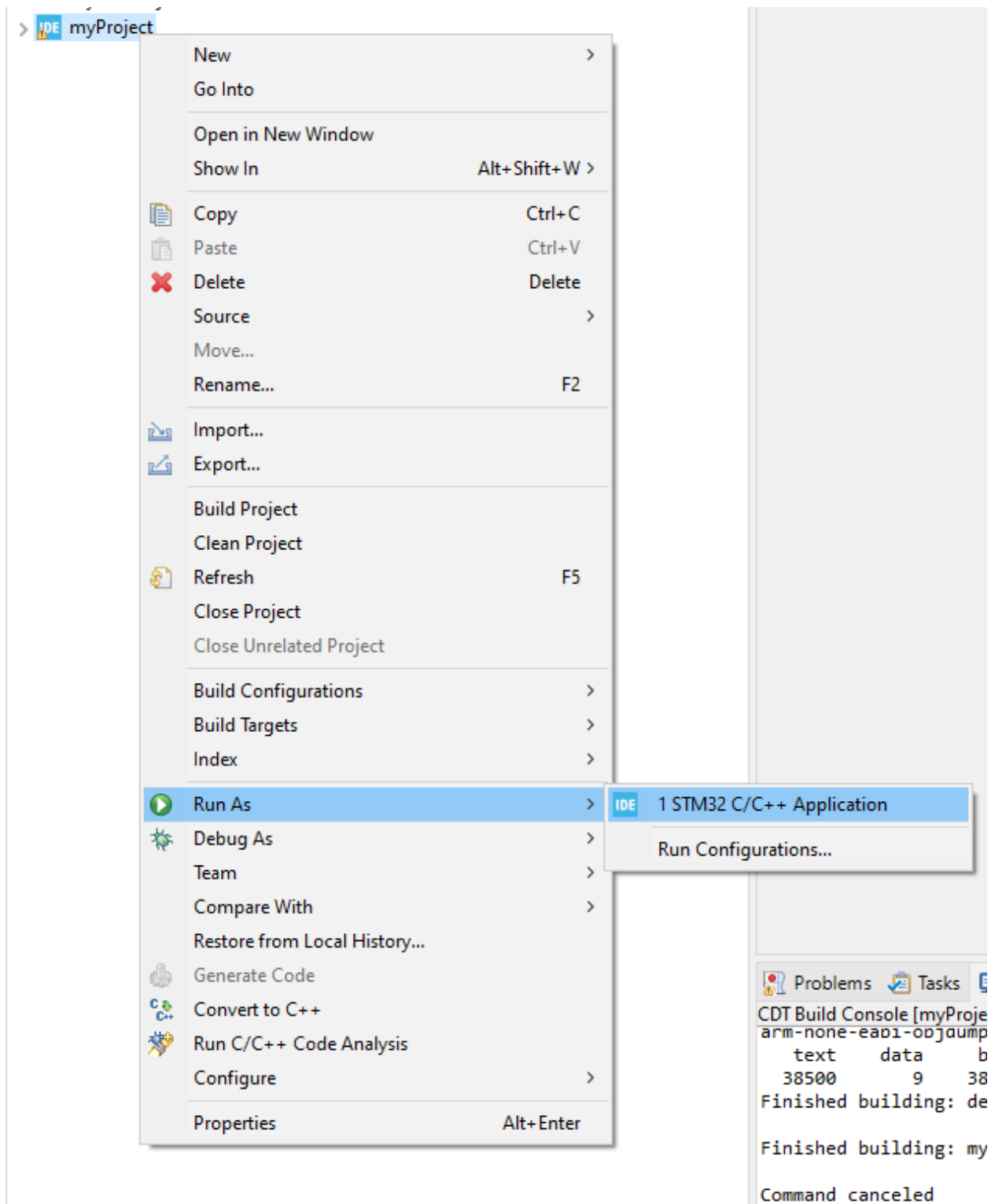
To generate the project in STM32CubeIDE:

- Click on **Generate Code**.



- Click on **Open Project** to open the project with STM32CubeIDE.

- 9) STM32CubeIDE will open. If STM32CubeIDE asks, select the default workspace. You should now see myProject on the left. Right-click on it and select Run As STM32 c/c++ application:



- 10) You should see many LEDs on your board, two of them (one red one green) next to each other. Lets make them flashing ☺ Find myProject->Core->Src->main.c and at line 136 the following while loop:

```
136 while (1)
137 {
138     /* USER CODE END WHILE */
139
140     /* USER CODE BEGIN 3 */
141 }
142 /* USER CODE END 3 */
143 }
```

Below /* USER CODE BEGIN 3 */ paste the following code:

```
HAL_GPIO_TogglePin (GPIOH, GPIO_PIN_6);
HAL_Delay (500);
HAL_GPIO_TogglePin (GPIOH, GPIO_PIN_7);
HAL_Delay (500);
```

Press Run and wait for the program to be compiled and transferred to the board. Did anything change?